PopCNVBrowser: a web-based Platform for CNV analysis and visualization using 1000 Genomes Project Data

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Introduction

- Copy number variants (CNVs) are crucial for understanding genetic diversity and disease susceptibility.
- The 1000 Genomes Project provides a foundation to study human genetic diversity.
- Effective comparison and visualization of CNV data across populations are vital to understand ethnic-specific patterns.

Objective

- We propose **PopCNVBrowser,** a web-based platform that not only offers powerful visualization tools for comparing results from various CNV calling methods but also serves as a comprehensive resource.
- This includes population-specific CNV frequencies and associations with disease susceptibility.

Materials

- **1000 Genomes Project:** This study uses data from the 1000 Genomes 30x on GRCh38 across 26 global populations covered 3202 samples.
- **DGV:** To ensure adherence to the CNV definition of 1Kbp or longer, we filtered out CNVs shorter than 1Kbp from the DGV gold standard Variants (GRCh38) data.
- **Ensemble Database:** We use Ensembl Genomes data (GRCh38.111) provided positional and functional information for genes.
- **ClinVar:** It was used to analyze and visualize the clinical significance and pathological associations of specific genetic variants

Methods

CNV calling methods

We used four CNV calling tools to analyze whole genome sequencing (WGS) data. For cn.MOPS, CNVKIT, and Control-FREEC, we performed analyses at multiple resolutions (5K, 25K, 50K) to capture CNVs of different lengths. However, readDepth does not allow for resolution adjustments, so only a single resolution result is provided per dataset.

Merge methods

To facilitate comparison across results, copy number regions (CNRs) are standardized using three approaches:

- **CNV Tool Merge**: Consolidates CNRs across the four for a single sample.
- **Sample Merge**: Merges CNR results for each sample within a population.
- **All Merge**: Integrates CNR results across tools and samples within each population.

Various analysis options are provided depending on CNV types (DUP/DEL) and different overlap ratios **(RO) from 0.5 to 0.9.**

Results

- Our platform provides three type of visualizations **: (1) Circular Bar Chart, (2) Linear Track, and (3) Navigation Table**
- Two key insights:
  (1) CNV distribution and copy number distribution
  (2) Disease associated regions

1. CNV Distribution Overview (Circular Bar Chart)

Genome Overview	Displays all chromosomes, allowing to select a specific chromosome.
Region Navigation	Enables zoom in, out, and panning of specific regions, synchronized with the linear track.
Bar Chart of CNV	Shows the average copy number within dynamic window sizes (10K-500K), representing duplications (green) or deletions (red).

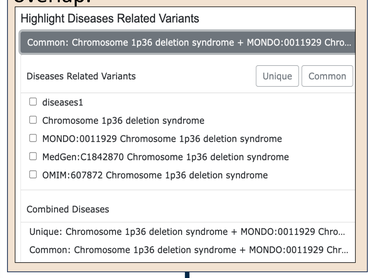


Hovering over shows details of bar

The number of CNVs in each chromosome

Pins are representing **the location of the CNV**

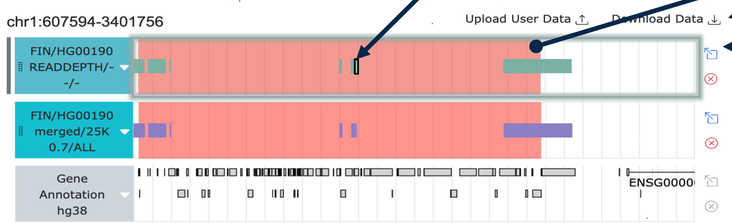
Selection of Disease
Users can toggle unique and common CNVs for each disease, with darker colors indicating more overlap.



Highlighting
Disease-related CNV regions

2. Population-Wide CNV Comparison (Linear Track)

- Allows comparison of CNV calls across different populations and resolutions, and merge approaches, including gene annotation.



Personal data can be added

Users can add, remove, or reorder results from four tools across multiple resolutions and merged results with different RO values.

3. Gene Selection for Enrichment Analysis (Navigation Table)

- Displays all genes within the selected chromosome, with CNVs marked as duplication (DUP) or deletion (DEL).

Query	Name	Start	End	Link	Overlap CNV type
☑	DDX11L2	11869	14409	GeneCards	DUP
☑	DDX11L1	12010	13670	GeneCards	DUP
☑	WASH7P	14696	24886	GeneCards	DUP
☑	FAM138A	34554	36081	GeneCards	DUP
☐	OR4G4P	52473	53312	GeneCards	-
☐	OR4G11P	62949	63887	GeneCards	-

Quick Select — Selected genes: 207

All CNV | All DUP | All DEL | Clear — Execute g:GOst with selected genes. Run g:GOSt

Conclusion

- Our platform provides insights into CNV functions across populations, enabling cross-ethnic comparisons and understanding disease genetics.
- Currently, it supports only 1000 Genomes data but can be extended to other datasets like UK Biobank.

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Reference

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